

# Retail Sales Analysis and Visualization Report

## 1. Project Overview

The objective of this project is to analyze retail sales transactions using Python and Power BI. The analysis focuses on understanding transaction patterns, revenue trends, product category performance, and customer purchasing behavior. The cleaned dataset was processed using Python and visualized using Power BI through interactive dashboards.

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## 2. Data Preparation

The following preprocessing steps were performed using Python:

- Loaded the dataset using Pandas.
- Converted the Date column into datetime format.
- Created a Revenue column using:

Revenue = Price × Quantity

- Extracted Month information from the Date column.
  - Handled missing values in CustomerNo.
  - Checked for Duplicate records.
  - Exploratory Data Analysis (EDA) done.
  - Visualized necessary plots in jupyter notebook from insights.
  - Exported the cleaned dataset for Power BI visualization.
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## 3. Executive Dashboard Analysis

### 3.1 Total Revenue

#### Observation

The dashboard reports a total revenue of 111.18 Million.

## **Interpretation**

The business generated substantial sales revenue across all products and customer segments. Revenue serves as the primary performance indicator and reflects the overall success of business operations.

## **Business Insight**

A high revenue base indicates strong product demand and effective sales performance across multiple categories.

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## **3.2 Total Customers**

### **Observation**

Approximately 90,000 unique customers participated in transactions.

### **Interpretation**

The organization has a broad customer base with significant market reach.

### **Business Insight**

A large customer base reduces dependency on individual customers and supports long-term business sustainability.

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## **3.3 Total Orders**

### **Observation**

The dataset contains approximately 375,000 orders.

### **Interpretation**

Customers have made multiple purchases, indicating repeat buying behavior.

### **Business Insight**

The high order count demonstrates consistent customer engagement and purchasing activity.

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## 3.4 Average Order Value (AOV)

### Observation

The Average Order Value is approximately 296.18.

### Interpretation

On average, customers spend around 296 units per transaction.

### Business Insight

Increasing the Average Order Value through cross-selling, bundling, and promotional strategies can significantly improve overall revenue.

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## 3.5 Revenue Trend Over Time

### Observation

Monthly revenue remains relatively stable throughout the year, fluctuating between approximately 9 Million and 10 Million.

### Interpretation

Revenue shows consistent performance with minor seasonal variations. February records the lowest revenue while stronger performance is observed during January, July, October, and December.

### Business Insight

The business demonstrates stable revenue generation and is not heavily dependent on a single period of the year.

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## 3.6 Revenue by Category

### Observation

Home Accents generates the highest revenue, followed by Decor Essentials and Gifting & Occasions.

### **Interpretation**

Customers show a stronger preference for Home Accents products compared to other categories.

### **Business Insight**

Home Accents should be prioritized for inventory management, promotions, and marketing investments due to its significant revenue contribution.

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## **3.7 Customer Distribution by Category**

### **Observation**

Customer distribution across categories is relatively balanced.

### **Interpretation**

Each category attracts a similar proportion of customers.

### **Business Insight**

Revenue differences among categories are likely driven by customer spending patterns rather than customer volume.

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## **3.8 Distribution of Transaction Amounts**

### **Observation**

Most transactions occur within lower revenue ranges, while higher transaction amounts occur less frequently.

### **Interpretation**

The transaction amount distribution is positively skewed (right-skewed). A large number of customers make smaller purchases, while a small number of customers generate high-value transactions.

### **Business Insight**

High-value transactions contribute significantly to total revenue despite occurring less frequently.

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## **3.9 Top 10 Products**

### **Observation**

Products such as Glass Bowl Set, Ceramic Vase, Children's Toy, and Vintage Clock generate the highest revenue.

### **Interpretation**

A limited number of products contribute a substantial share of overall sales.

### **Business Insight**

These products should receive priority in inventory planning, advertising campaigns, and promotional activities.

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# **4. Customer Analytics Dashboard Analysis**

## **4.1 Monthly Active Customers**

### **Observation**

Monthly active customers remain stable throughout the analysis period.

### **Interpretation**

Customer participation remains consistent with only minor fluctuations.

## **Business Insight**

The business demonstrates strong customer retention and recurring purchasing behavior.

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## **4.2 Monthly Revenue**

### **Observation**

Monthly revenue follows a relatively stable pattern with moderate fluctuations.

### **Interpretation**

Revenue performance remains predictable and consistent across months.

## **Business Insight**

Stable revenue trends improve forecasting accuracy and strategic planning.

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## **4.3 Customers by Country**

### **Observation**

Customers are distributed across multiple countries with relatively balanced representation.

### **Interpretation**

The customer base is geographically diversified.

## **Business Insight**

The business is not overly dependent on a single geographic market, reducing market concentration risk.

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## **4.4 Geographic Distribution of Customers**

### **Observation**

Customer activity is concentrated primarily in Europe, North America, and Australia.

### **Interpretation**

These regions represent the strongest markets for the business.

### **Business Insight**

Future expansion opportunities may exist in underrepresented regions such as Asia, Africa, and South America.

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## **4.5 Top 10 Customers**

### **Observation**

The highest-value customers contribute similar levels of revenue.

### **Interpretation**

Revenue is distributed across multiple customers rather than being concentrated among a small group.

### **Business Insight**

Low customer concentration risk enhances business stability and reduces dependence on individual customers.

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## **5. Answers to Assignment Questions**

### **a. What is the distribution of transaction amounts?**

The transaction amount distribution is positively skewed. Most transactions occur in lower revenue ranges, while high-value transactions occur less frequently.

### **b. How does the total transaction amount vary over time?**

Total revenue remains relatively stable throughout the year with minor fluctuations, indicating consistent sales performance.

**c. Which product categories generate the most revenue?**

Home Accents generates the highest revenue, followed by Decor Essentials and Gifting & Occasions.

**d. Can you identify any trends or patterns in customer purchasing behaviour?**

The Monthly Active Customers chart shows that customer activity remains relatively stable throughout the year, generally ranging between approximately 16,500 and 18,500 active customers. However, a noticeable decline in customer activity is observed during **February**, followed by a recovery in subsequent months.

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## **6. Conclusion**

The retail sales analysis reveals a healthy business with strong revenue performance, a diversified customer base, and stable purchasing behavior. Home Accents emerges as the leading revenue-generating category, while transaction patterns indicate that smaller purchases occur more frequently than high-value transactions. The business demonstrates consistent customer engagement, balanced geographic distribution, and sustainable growth potential.